Credit Card Fraud Detection using Deep Clustering and Neural Networks

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ABSTRACT

As digital payments continue to grow, incidents of credit card and identity fraud have become increasingly common. Many individuals, including close family members, have experienced unauthorized purchases or account misuse, highlighting the urgent need for effective fraud detection systems. Artificial intelligence and neural networks offer a promising avenue to combat these threats by automating detection and identifying complex fraud patterns that traditional methods often miss. Having a good credit score is crucial for major life decisions such as home purchases and employment opportunities; therefore, ensuring financial security is of societal importance.

This project leverages deep neural networks to enhance credit card fraud detection, addressing challenges such as data imbalance and evolving fraud behaviors. By integrating Synthetic Minority Oversampling Technique (SMOTE) and deep clustering with neural architectures, the goal is to improve both detection accuracy and sensitivity to minority (fraudulent) cases. The system aims to achieve reliable fraud detection performance while maintaining efficiency suitable for real-world financial applications.

Keywords: Credit card fraud detection, Deep Clustering, Neural Networks, SMOTE, Imbalanced data, Anomaly detection, Precision, F1-score

INTRODUCTION

Using the Kaggle Credit Card Fraud Detection dataset as a source, this research focuses on developing a deep neural network-based approach to identify fraudulent credit card transactions. The dataset comprises 284,807 transactions with only 492 fraudulent cases (0.172% fraud rate), presenting a significant class imbalance challenge that requires sophisticated neural network architectures [1]. Detecting fraudulent activity in such imbalanced data requires advanced algorithms capable of recognizing subtle patterns that differentiate legitimate and fraudulent transactions.

Recent developments in deep learning have shown promising results in financial fraud detection, with neural network approaches achieving superior performance compared to traditional machine learning methods [2][5]. Neural networks can automatically learn hierarchical and non-linear relationships from raw data, reducing the need for manual feature engineering. To handle class imbalance and improve minority-class recognition, oversampling techniques such as Synthetic Minority Oversampling Technique (SMOTE) are often used. However, while SMOTE can improve recall by generating synthetic fraud samples, it may also reduce precision or lead to overfitting when applied to highly imbalanced data.

This project introduces a hybrid approach integrating Deep Clustering with Neural Networks to uncover latent transaction behavior patterns and improve detection sensitivity. By clustering transactions into behaviorally similar groups using K-Means and training specialized neural networks within each cluster, the model aims to capture diverse fraud characteristics. The combination of oversampling-based learning and cluster-specific modeling is expected to enhance precision, recall, and adaptability to evolving fraud patterns.

RESEARCH STATEMENT

The primary objective of this research is to design an effective and adaptive neural network framework for detecting credit card fraud in highly imbalanced datasets. The study aims to improve both the sensitivity and precision of fraud detection while maintaining computational efficiency suitable for real-world deployment.

This work explores the integration of two complementary strategies: SMOTE-based oversampling for addressing extreme class imbalance and deep clustering for capturing hidden behavioral structures within transactions. The combined approach is expected to enhance detection accuracy by enabling the model to generalize across diverse fraud patterns rather than relying on a single global classifier.

Unlike previous studies that either oversample the minority class or employ a single deep model for all transactions, this research emphasizes a hybrid framework where clustering supports behavioral segmentation, and each cluster-specific neural network focuses on distinct transaction characteristics. This design introduces a novel perspective on adaptive fraud analytics, improving robustness against evolving fraud trends and providing a foundation for interpretable and efficient fraud detection systems.

HYPOTHESIS

We propose two key hypotheses for our credit card fraud detection project:

1. Baseline SMOTE-Augmented Neural Network Hypothesis: Applying the Synthetic Minority Over-sampling Technique (SMOTE) to balance the training data before training a multi-layer perceptron will significantly improve fraud detection performance (measured by F1-score) compared to training the same network on the raw, imbalanced dataset.

2. Deep Clustering with Neural Networks Hypothesis: Partitioning transactions into behaviorally similar clusters using K-Means and training separate neural network models for each cluster will uncover latent patterns and yield higher overall detection sensitivity and specificity (measured by AUC-ROC and recall) than a single model trained on the entire dataset.

PRELIMINARY LITERATURE SURVEY

Existing research on credit card fraud detection demonstrates a clear progression from traditional machine learning models toward hybrid and deep learning frameworks. One of the earliest works, "Enhancing Credit Card Fraud Detection: A Neural Network and SMOTE Integrated Approach" [1], established the effectiveness of applying the Synthetic Minority Oversampling Technique (SMOTE) to balance the extremely imbalanced European credit card dataset (0.172% fraud rate). By combining SMOTE with a multi-layer neural network, the study achieved near-perfect precision and recall, demonstrating that data resampling significantly enhances minority class detection compared to classical models such as SVM and Decision Trees.

Building on this foundation, "CTCN: A Novel Credit Card Fraud Detection Method Based on Conditional Tabular GANs and Temporal Convolutional Networks" [2] introduced a temporal dimension to fraud analysis. By integrating CTGAN-generated synthetic samples with a Temporal Convolutional Network architecture, the study captured long-range dependencies in transaction sequences more effectively than LSTM-based models. This work underscores the potential of combining generative and temporal architectures for improved fraud modeling.

Further improvements were explored in "A New Fusion Neural Network Model and Credit Card Fraud Identification" [3], where deep and wide neural components were fused to learn both complex hierarchical features and direct feature interactions. This joint learning strategy improved classification accuracy and reduced inference time, suggesting that balanced model design can enhance both interpretability and performance.

In parallel, "Securing Transactions: A Hybrid Dependable Ensemble Machine Learning Model Using IHT-LR and Grid Search" [4] emphasized ensemble optimization. By integrating Decision Tree, Random Forest, KNN, and Multi-layer Perceptron classifiers, and tuning weight assignments using grid search, the model achieved exceptional accuracy and near-perfect ROC scores. This highlights the strength of adaptive weighting and hybrid ensemble design for imbalanced problems.

A broader systematic analysis was presented in "Year-over-Year Developments in Financial Fraud Detection via Deep Learning: A Systematic Literature Review" [5]. The authors synthesized several state-of-the-art techniques, including SMOTE, ADASYN, GAN-based data synthesis, and deep ensemble strategies, reinforcing that hybrid deep learning approaches consistently outperform single-model methods. The paper also emphasized using PR-AUC as a more stable measure for imbalanced evaluation.

Finally, "The Advantages of the Matthews Correlation Coefficient (MCC) over F1-score and Accuracy in Binary Classification Evaluation" [6] addressed evaluation reliability. The authors demonstrated that MCC provides a more balanced assessment than accuracy or F1 when classes are skewed, as it equally penalizes false positives and false negatives. For fraud detection, this metric offers a more truthful reflection of classifier behavior.

Collectively, these studies form the foundation for the present research. Works [1]–[5] establish deep neural networks and hybrid learning as the most effective direction for fraud analytics, while [6] informs the evaluation framework by introducing MCC as a reliable complement to F1 and AUC. Together, they validate the proposed approach of combining SMOTE-based balancing with deep clustering and specialized neural networks to address the dual challenges of class imbalance and behavioral diversity in credit card transactions.

METHODOLOGY

The methodology will outline the systematic plan for developing a neural network–based fraud detection framework that effectively addresses extreme class imbalance and hidden behavioral diversity in transactions. The overall approach will combine oversampling-based learning with deep clustering–assisted specialization to improve detection precision, recall, and adaptability to evolving fraud patterns. All experiments will be implemented using Python in Google Colab, with TensorFlow, scikit-learn, and imbalanced-learn libraries for modeling, preprocessing, and resampling.

1. Data Collection:

Approach: The Kaggle Credit Card Fraud Detection dataset will be used for all experiments. It contains 284,807 transactions, out of which only 492 are fraudulent, yielding a fraud rate of 0.172%. This dataset is widely cited in prior research and provides a standard benchmark for evaluating fraud detection methods under severe imbalance.

Details: Each record consists of 30 numerical features, including 28 anonymized components (V1–V28) derived from Principal Component Analysis (PCA), along with Time, Amount, and Class. These PCA features will serve as reduced-dimensional representations that retain most of the original variance, thereby minimizing redundancy and ensuring computational efficiency for subsequent clustering. The chronological order of transactions will be preserved to simulate real-world transaction flow. The dataset will be imported into the Colab environment for preprocessing, model development, and evaluation.

2. Data Preprocessing and Feature Engineering:

Approach: Preprocessing will standardize and normalize features, address data imbalance, and prepare the dataset for neural network training and clustering-based modeling.

Details: The Time and Amount features will be standardized using z-score scaling to ensure equal contribution to model training. The PCA-transformed components (V1–V28) will be normalized to maintain consistent magnitude across features. The data will be split chronologically into 70% training, 15% validation, and 15% test sets, ensuring that later transactions do not leak into earlier phases. The Synthetic Minority Oversampling Technique (SMOTE) will be applied exclusively on the training set to generate synthetic fraud instances while maintaining the natural data distribution in validation and test sets. This setup will help the neural network learn more representative decision boundaries for minority transactions without distorting evaluation integrity.

PCA will also play a critical role in reducing dimensional complexity before clustering, enabling the subsequent Deep Clustering stage to identify meaningful transaction groups efficiently.

3. Baseline Neural Network with SMOTE:

Approach: The first experimental phase will establish a baseline neural network trained on the SMOTE-balanced dataset. This model will quantify how oversampling influences minority detection performance and will serve as the foundation for subsequent experimentation.

Details: The baseline model will consist of a Multilayer Perceptron (MLP) with three hidden layers (64, 32, and 16 neurons), each using ReLU activation functions, followed by a sigmoid output layer for binary classification. The network will be trained using binary cross-entropy loss and optimized with the Adam optimizer. Early stopping based on validation F1-score will be implemented to prevent overfitting. The trained model will be evaluated on the test dataset using Precision, Recall, F1-score, AUC-ROC, and MCC to ensure a fair assessment of performance under imbalance.

This baseline will confirm the impact of SMOTE on recall and AUC improvement and will serve as a benchmark for the planned hybrid approach.

4. Deep Clustering + Neural Networks:

Approach: To enhance adaptability and precision, the next phase will introduce a Deep Clustering + Neural Network hybrid, representing the novel contribution of this study. This hybrid design will combine unsupervised clustering and supervised classification to detect subtle variations in fraud behavior that a single global model might overlook.

Details: K-Means clustering will be applied to the PCA-normalized dataset to group transactions into clusters based on behavioral similarity. Different values of k (2, 4, 5, 10, and 20) will be explored to evaluate model scalability and specialization. Each cluster will undergo an independent 70/15/15 chronological split, and the baseline MLP architecture will be retrained separately on each cluster. If class imbalance persists within a cluster, weighting strategies or targeted oversampling will be applied locally.

At inference time, a new transaction will first be assigned to its nearest cluster centroid, and the corresponding trained cluster-specific neural network will generate its prediction. The ensemble of these specialized networks will collectively determine the final classification. This hierarchical structure is expected to improve minority detection by focusing on cluster-level variations while maintaining computational efficiency and interpretability.

5. Hyperparameter Optimization:

Approach: Optimization will ensure that the model generalizes well across varying cluster sizes and data distributions. Parameters influencing learning dynamics and regularization will be tuned to achieve balanced recall and precision.

Details: Learning rates will be explored within the range of 0.001 to 0.00001 using decay schedules where necessary. Batch sizes will vary between 32 and 128 to control convergence stability. Regularization will include dropout (0.2–0.4) and L2 penalties to reduce overfitting, and early stopping will monitor validation AUC for convergence control. Manual tuning and grid search will be employed to select the optimal configuration that maximizes recall, F1-score, and PR-AUC while maintaining consistent AUC-ROC and MCC.

6. Evaluation Framework:

Approach: Evaluation will employ multiple performance measures to capture both overall discrimination and minority-class behavior, addressing limitations of single-metric reporting in imbalanced contexts.

Details: The F1-score will provide a harmonic balance between precision and recall, while AUC-ROC will measure global separability between classes. PR-AUC will further evaluate precision–recall performance under class imbalance. Precision and Recall will be analyzed individually to understand trade-offs between false positives and false negatives. Finally, Matthews Correlation Coefficient (MCC), recommended in prior studies [6], will be included as a reliable indicator of binary performance on imbalanced datasets. These metrics together will offer a comprehensive assessment of model quality, stability, and fairness across both phases of experimentation.

EXPERIMENTAL DESIGN

1. Dataset Acquisition and Setup:

The Kaggle Credit Card Fraud Detection dataset was selected as the benchmark dataset for this study. It contains 284,807 transaction records with 492 confirmed frauds, resulting in an imbalance ratio of 0.172%. The dataset was downloaded from Kaggle and imported into a shared Google Drive workspace, which was then mounted to Google Colab for preprocessing, model

training, and evaluation. This centralized setup ensured consistent access across all project members and reproducibility of results.

2. Environment Configuration:

All experiments were conducted in a Google Colab environment configured with Python 3.10, TensorFlow, scikit-learn, and imbalanced-learn libraries. GPU acceleration was used when available; otherwise, the notebook automatically reverted to CPU execution. Random seeds were fixed to maintain consistency across runs. The environment setup allowed seamless execution of both preprocessing and model evaluation workflows within a single controlled notebook.

3. Data Loading and Preprocessing:

The full dataset was loaded into a Pandas DataFrame and verified for missing values and feature consistency. The Time and Amount features were standardized using z-score normalization, while the PCA-transformed components (V1–V28) were scaled to a uniform numerical range. The data was chronologically divided into 70% training, 15% validation, and 15% test subsets to preserve real-world temporal ordering.

To address the extreme class imbalance, the Synthetic Minority Oversampling Technique (SMOTE) was applied exclusively to the training set, generating synthetic fraud samples while keeping the validation and test distributions untouched. The Class attribute was encoded as a binary label: 0 for legitimate and 1 for fraudulent transactions.

4. Model Implementation:

The Baseline Neural Network with SMOTE served as the first experimental setup. The model architecture consisted of three hidden layers (64, 32, and 16 neurons) with ReLU activations and a sigmoid output neuron for binary classification. Training was conducted using binary cross-entropy loss and the Adam optimizer, with early stopping based on validation AUC to prevent overfitting.

This experiment aimed to establish a reliable reference model for imbalanced classification before proceeding to the upcoming Deep Clustering + Neural Network hybrid in the next phase. The baseline implementation was validated and evaluated using the metrics Precision, Recall, F1-score, AUC-ROC, and MCC.

RESULTS AND ANALYSES

1. Baseline Experiment — Neural Network + SMOTE:

The primary objective of the baseline experiment was to evaluate how Synthetic Minority Over-sampling Technique (SMOTE) affects fraud detection performance when integrated with a multilayer neural network trained on an extremely imbalanced dataset.

1.1 Model Training Performance:

Training was conducted for 30 epochs using binary cross-entropy loss and the Adam optimizer. Early stopping was enabled based on validation AUC. The model showed rapid convergence within the first 10 epochs, achieving a stable validation AUC of 0.9681 at epoch 6, indicating effective generalization to unseen data despite oversampled inputs.

Metric	Validation Set
Best Validation AUC	0.9681
Final AUC (after 30 epochs)	0.978
Validation Loss Trend	Decreasing steadily, showing no overfitting

Table 1: *Training and validation performance metrics for the baseline NN + SMOTE model*

1.2 Validation Results:

Metric	Value
AUC	0.9781
Precision	0.2025
Recall	0.8571
F1-Score	0.3276

Table 2: *Validation set evaluation metrics for the baseline NN + SMOTE model*

	Predicted 0	Predicted 1
Actual 0	42 476	189
Actual 1	8	48

Table 3: *Confusion matrix showing classification outcomes on the validation set*

The model correctly identified 85.7 % of fraudulent transactions while maintaining high specificity for legitimate cases. The moderate F1-score (≈ 0.33) reflects the expected trade-off caused by limited precision, a known outcome when recall is prioritized in high-imbalance contexts. These metrics confirm that SMOTE helped the model detect far more fraud cases than an unbalanced baseline would have achieved.

1.3 Test Results:

Metric	Value
AUC	0.9513
Precision	0.2500
Recall	0.8077
F1-Score	0.3818

Table 4: *Test set evaluation metrics for the baseline NN + SMOTE model*

	Predicted 0	Predicted 1
Actual 0	42 544	126
Actual 1	10	42

Table 5: *Confusion matrix showing classification outcomes on the test set*

Performance on the held-out test data remained strong, achieving a test AUC of 0.9513 and recall of 0.8077. The slight decline from validation values indicates normal generalization loss rather than overfitting. The improved F1-score (≈ 0.38) and increased precision demonstrate that the model retained its ability to identify fraudulent transactions while reducing false positives compared to the validation stage.

1.4 Overall Analysis:

- **AUC Improvement:** Achieving AUC ≈ 0.95–0.98 confirms that the neural network captured non-linear relationships between PCA-reduced features effectively.
- **Recall Strength:** Recall ≈ 0.85 indicates that most fraudulent samples were detected, an essential property for fraud-prevention systems where missing a fraud is costlier than flagging a legitimate transaction.
- **F1 Limitation:** Despite strong recall, F1-scores remained around 0.33–0.38 because precision is still low (≈ 0.20–0.25). This limitation arises from synthetic oversampling, which can slightly inflate the minority class boundary and cause more false positives.
- **Model Reliability:** The close alignment of validation and test AUC values demonstrates that the baseline model generalizes well without significant variance, verifying the effectiveness of chronological splitting and SMOTE isolation to prevent data leakage.

2. Second Phase — Deep Clustering + Neural Network:

The main goal of this experiment was to see how well a deep autoencoder, when paired with K-means clustering and separate neural network models per cluster, can improve fraud detection on a highly imbalanced dataset. Instead of relying on oversampling methods like SMOTE, this approach tries to learn hidden transaction patterns by compressing the data into a latent space and then training specialized models for each behavioral group. The idea is that clustering in the autoencoder space should isolate different transaction profiles, allowing each neural network to focus on a more consistent subset of the data and potentially detect fraud more effectively.

2.1 Model Training Performance:

The autoencoder was trained on 170,884 samples using binary cross-entropy and the Adam optimizer, and it converged quickly without instability. The latent space it learned remained consistent across epochs, which helped produce stable cluster assignments. After that, each cluster got its own neural network, trained with early stopping based on its own validation loss.

Across all settings k = 10 ended up being the most reliable and best performing configuration. At this level, clusters had enough fraud cases to train meaningful local models without becoming too sparse or noisy.

Metric	Value
Best Cluster AUC	0.9202
PR-AUC (overall)	0.7559
Fallback Model Use	Minimal use, only for sparse clusters
Validation Loss Trend	Stable with no overfitting observed

Table 1: *Training and validation performance for the deep-clustering model (k=10)*

The stable validation loss, along with the strong PR-AUC, shows that the autoencoder's latent space is working correctly for separating fraud from normal transactions. This separation is what allows the model to perform much better at moderate k values compared to the low-k setups, where more data gets mixed together.

2.2 Validation Results:

The aggregated validation metrics for k = 10 reflect the best balance between alert precision and fraud sensitivity.

Metric	Value
ROC AUC	0.9202
Precision	0.3563
Recall	0.8267
F1-Score	0.4980

Table 2: *Validation metrics for the deep clustering model (k=10)*

	Predicted 0	Predicted 1
Actual 0	56,775	112
Actual 1	13	62

Table 3: *Confusion matrix for validation samples (k=10).*

These results show that the model caught about 82.7% of all fraud cases while keeping false positives much lower than in the smaller k setups. The F1-score of roughly 0.50 reflects a solid balance between precision and recall, and it clearly outperforms both the low-k models that over-trigger, and the high-k models that become too sparse to learn anything useful.

2.3 Test Results:

The final model was tested across all clustering settings from k = 2 up to k = 50 to see how different levels of segmentation influence fraud-detection performance. The consolidated metrics are listed in Table 4, covering ROC AUC, PR AUC, precision, recall, and F1-score for each configuration.

k	ROC AUC	PR AUC	Precision	Recall	F1-Score
2	0.9505	0.6908	0.0305	0.8800	0.0590
4	0.9341	0.6926	0.0243	0.8800	0.0473
5	0.9030	0.4494	0.0194	0.8400	0.0381
10	0.9201	0.7558	0.3563	0.8267	0.4979
20	0.9462	0.7136	0.1358	0.8400	0.2338
50	0.9456	0.6801	0.1161	0.7200	0.2000

Table 4: *Test-set performance metrics from all clustering configurations.*

The trend shows that when k is small (2–5), recall goes up but precision collapses, which shows that the model is flagging many normal transactions because the clusters are still too broad. On the other end, pushing k too high (20–50) breaks the data into very small and sparse groups that most clusters don't have enough fraud to train anything meaningful, making performance unstable. At k = 10 the model reaches its best balance between catching fraud and keeping false alarms under control. With an F1-score near 0.50 and the strongest PR-AUC (~0.756), k = 10 captures useful structure in the latent space without over-segmenting the data.

2.4 Overall Analysis:

- **Best overall performance at k = 10:** k = 10 consistently delivered the strongest results, with the highest F1-score (0.498) and the top PR-AUC (≈0.756). This suggests that around ten behavioral clusters is the point where the latent space structure is captured cleanly without breaking the data apart too aggressively.
- **Low-k models hit high recall but collapse on precision:** For k = 2, 4, and 5, recall stayed above 0.84, but precision dropped below 0.03. This happens because small-k settings create very broad clusters that mix different transaction patterns, causing a surge of false alarms.
- **High-k models (20–50) become too sparse to learn anything stable:** When k gets large, many clusters end up with only a handful of fraud cases. This forces more fallback predictions and causes inconsistent precision and weak F1-scores. The models simply don't have enough local data to learn reliable boundaries.
- **Autoencoder embeddings still separate fraud well:** Even when F1 dropped at extreme k values, ROC-AUC stayed high (≈0.93–0.95). This shows that the latent space does preserve good separability, but the per-cluster classifiers need reasonable cluster sizes to use that structure effectively.
- **PR-AUC is the metric that reflects real performance:** While ROC-AUC stays high across all k, PR-AUC changes dramatically with cluster quality. It's the metric that matches what matters in fraud detection: balancing true positives without flooding the system with false alarms.
- **Deep clustering enables more specialized and adaptive fraud detectors:** The cluster-specific neural networks learned localized decision patterns tied to each behavioral group. This gives a clear advantage over a single global model or simple oversampling, since the decision boundaries adapt to the structure discovered in the autoencoder's latent space.

3. Progress Toward Next Phase:

With the Deep Clustering + Neural Network framework now operational, the next major step is to refine the system through targeted hyperparameter optimization. Since the dataset is highly imbalanced and the cluster-specific models are sensitive to overfitting than global models, tuning these parameters will be a critical part of improving performance. The goal is to tighten training stability inside each cluster while pushing precision, recall, and F1-score beyond the current baseline. This stage will directly determine how well the system generalizes rare fraud behaviors and how effectively it can reduce false positives without sacrificing detection sensitivity.

TO-DO LIST AND FUTURE WORK

The following tasks outline the upcoming work toward enhancing performance through Hyperparameter Optimization.

- **Perform hyperparameter tuning within clusters:** Optimize key parameters including learning rate, batch size, dropout rate, and weight decay for each cluster model. The tuning will aim to maximize F1-score and PR-AUC while maintaining stable recall performance and preventing overfitting.
- **Evaluate and compare results:** Compute Precision, Recall, F1-Score, AUC-ROC, PR-AUC, and MCC for each cluster-specific model. Compare these outcomes with the baseline NN + SMOTE results to assess improvements in fraud-detection precision and stability.
- **Integrate ensemble prediction:** Combine the outputs from all cluster-specific networks into a unified ensemble model that aggregates probabilities for the final classification decision. This integration is expected to enhance overall robustness and reduce false positives.
- **Visualize and document findings:** Prepare comparative charts and performance summaries illustrating the effect of clustering on detection accuracy. Consolidate observations and analyses for inclusion in the next phase of reporting and manuscript preparation.

PROJECT MANAGEMENT

Team Roles:

- Sukumar Muthusamy – Baseline Neural Network + SMOTE implementation, model evaluation, and documentation.
- Mauricio Gonzalez – Deep Clustering + Neural Network development, clustering validation, and hybrid model integration.
- Dhruv Shrivastava – Hyperparameter tuning, optimization, and performance analysis.

Tools and Technologies:

- Development Environment: Google Colab (Python 3.10)
- Machine Learning Libraries: TensorFlow, scikit-learn, imbalanced-learn
- Data Handling and Analysis: Pandas, NumPy
- Visualization Libraries: Matplotlib, Seaborn, Plotly

- Collaboration Platform: Microsoft Teams and OneDrive

Backup Plan:

- All collaborative notebooks are saved and synchronized within Colab.
- Microsoft OneDrive Documents saved instantaneously.

Communication:

- Weekly check-ins via Microsoft Teams meeting.
- Daily updates through Microsoft Teams messenger.

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